

YERDOS NARZHIGITOV

github.com/YerdosNar

PROFILE

Strong interest in low-level programming and networking (protocol obfuscation and encryption).
Also interested in UNIX-like environment (especially GNU/Linux).

CONTACT ME

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Gangwon-do,
Chuncheon-si,
Aemakgol-gil

EDUCATION

KANGWON NATIONAL UNIVERSITY
B.Sc. Computer Science and Engineering
2023-2027

LANGUAGE

Advanced English
Native Russian
Native Kazakh
Native Uzbek
Advanced Korean (TOPIK 6)

COMPUTER SKILLS

Git, GDB, ASAN, pThreads, LibSodium
C11, Bash, Java, Python
Linux (Arch, Debian, Alma)

WORK EXPERIENCE

OFFICE OF INTERNATIONAL AFFAIRS
2 years - International supporter at the Office of International Affairs at Kangwon National University.

CHUNCHEON HONORARY INTERPRETERS
1 year - Translator (English-Russian → Korean) at Chuncheon City Hall.

Engineering Projects

OpenP2P: E2EE Messenger/File-transfer & Proxy (C11, Makefile)

- Peer-to-peer communication tool featuring NAT hole-punching.
- End-to-end encryption and forward secrecy using X25519, XChaCha20-Poly1305.
- Authenticated handshake protocol to resist Man-in-the-Middle (MITM) attacks.
- POSIX threads to handle concurrent operations.
- Validated memory safety using AddressSanitizer (ASAN) and GDB.

MNIST Handwritten Digit Recognition Neural Network (C11)

- Feedforward operation, the backpropagation algorithm, and ReLU activation function through manual memory management in C with no libraries.
- Graphical 28x28 canvas (using SDL2) allows users to draw digits and test the neural network.

3X-UI Panel Automated Installation (Bash)

- Bash script (>1K LOC) to fully automate the installation of 3X-UI panel.
- Network stack optimization by modifying sysctl configuration file to enable TCP BBR congestion control, and expand TCP buffers, increasing VPN throughput to maximum.
- UFW for firewall, Caddy for reverse proxy and HTTPS/SSL certificate management.

Low-Level PNG Image Processor (C11, Makefile)

- Custom PNG parser from scratch capable of reading, decompressing (via zlib), and validating IHDR, IDAT, IEND chunks, alongside custom hidden chunks manipulation.
- Detects injected hidden data and can inject hidden data chunks (steganography).
- 3x3 convolution kernels to apply image processing filters (Sobel, Gaussian, Laplacian) via direct memory manipulation of pixel matrix.

UDP with ACK Simulation (C11, Bash)

- Simple Stop-and-Wait flow control over UDP with ACK packets.
- Simulates network instability, including packet loss, delays, and retransmission strategies.

Campfire website backend - 모닥불 (Team Project, Python, SQLite3)

- Collaborated in a 4-person agile team utilizing Git version control to build a real-time anonymous community backend.
- Engineered the core topic creation endpoints and implemented a similarity mapping algorithm to track and cluster trending discussions.
- No heavy ORMs to write a highly optimized Raw SQL queries for maximum DB retrieval performance.

TinyC (C11, Bash, Makefile) → Under development

- A very simple compiler that generates Assembly (Intel syntax) code from C source code.
- Supports 64-bit integers (8-byte), function calls, conditionals and loops.
- Under development.

Additional Information

Consistent Development Discipline

- Maintained a continuous 6-month daily coding streak (minimum 3 hours/day).
- Over 50 % of the codebase written in C and Bash.

Contributions to open source projects

- Blitz (Obfuscation Protocol: Hysteria Panel): Fixed the installation Bash script.
- Implemented process logging to streamline debugging for broken/interrupted deployments

MAT364 (Security Course of Suleyman Demirel University)

- Corrected the Bash script for Markdown-to-PDF conversion.
- Fixed calculation logic for encryption algorithm.

Server maintenance (VPS)

- Maintained a VPS with 3X-UI running in a docker.
- Added AdGuardHome to block adds by DNS.

Linux Familiarity

- Daily drive 2 Linux Distributions: Arch Linux and Void Linux.
- Main VPS running Debian Linux.
- Currently learning Alma Linux for RHCSA certificate.

PC Build Experience

- Have experience of working in a local PC building company.
- Familiar with hardware components.
- Experience of preventing bottlenecking by weak parts.